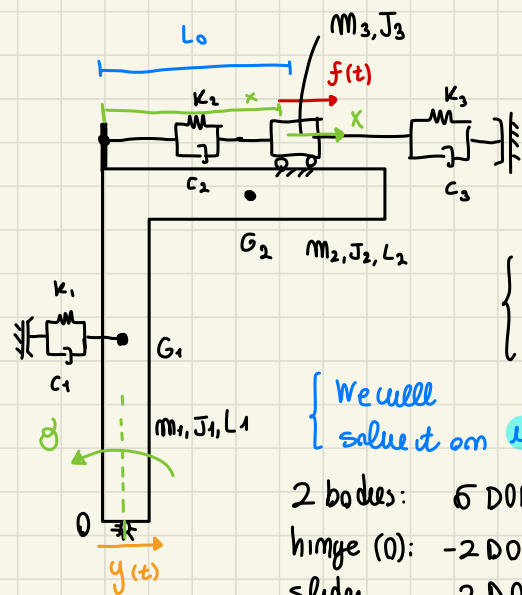




ESE IO EXAM PROBLEM 1



2 body syst { "L" shape providing separate J and m of the body part (as 2 separate bodies clamped)
 • slider, sliding over the upper part of the L body

{ constrain movement on the hinge on bottom y(t)
 external force applied on the slider f(t) }

{ We will solve it on MATLAB }

2 bodies: 6 DOF
 hinge (0): -2 DOF $\Rightarrow$ 2 DOF system
 slider -2 DOF easy to choose

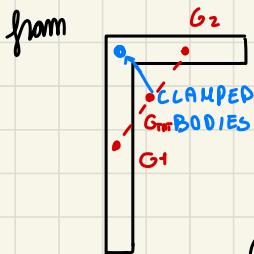
• JACOBBIAN MATRIX:

from mom lm relationship of all center of mass, than power series expansion to II ORD (simple power exp. cause $\theta = 0$)
 measure θ from vertical position, easy expansion!

↓
 from kinematic relationship approx at II ORD to identify $[A_m], \dots$

$$\underline{x} = \begin{pmatrix} \theta \\ \bar{x} \end{pmatrix}$$

relative displacements, variation of the position of the slider respect the corner
 x measure relative displacement // to the leg of the "L" body



it's easier to treat the overall body as union of two bodies with J_1, J_2
 But same rotation,

overall $m_1 + m_2 = m$ BUT in this way G_{TOT} will be on strange point out of body hand

↪ to express kinematic
 simpler to separate the body into different bodies!

simpler respect single body

↓ general displacement

We can neglect slider dimension



We consider G_3 as key here

$$\begin{cases} x_{G1} = -\frac{L_1}{2} \sin(\theta) + y \approx -\frac{L_1}{2} \theta + y \\ y_{G1} = \frac{L_1}{2} \cos(\theta) \approx \frac{L_1}{2} \left(1 - \frac{\theta^2}{2}\right) \end{cases} \quad (2^{nd} \text{ ORD terms})$$

$$\begin{cases} x_{G2} = -L_1 \sin(\theta) + \frac{L_2}{2} \cos(\theta) + y \approx -L_1 \theta + \frac{L_2}{2} \left(1 - \frac{\theta^2}{2}\right) + y \\ y_{G2} = +L_1 \cos(\theta) + \frac{L_2}{2} \sin(\theta) \approx L_1 \left(1 - \frac{\theta^2}{2}\right) + \frac{L_2}{2} \theta \end{cases}$$

general position,

AG_3 vary as $x + L_0$ parallel to bar

$$\begin{cases} x_{G3} = -L_1 \sin(\theta) + (L_0 + x) \cos(\theta) + y \approx -L_1 \theta + (L_0 + x) \left(1 - \frac{\theta^2}{2}\right) + y \\ y_{G3} = L_1 \cos(\theta) + (L_0 + x) \sin(\theta) \approx L_1 \left(1 - \frac{\theta^2}{2}\right) + (L_0 + x) \theta \end{cases}$$

↓
so from here we can find $[\lambda_m]$ overall general approach

3 rows & body (G_1, G_2 separate)

$$[\lambda_m] = \begin{bmatrix} \theta & \dot{x} & \dot{y} \\ -L_1/2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ -L_1 & 0 & 1 & 0 \\ L_2/2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ -L_1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} V_{G1x} \\ V_{G1y} \\ \omega_1 \\ V_{G2x} \\ V_{G2y} \\ \omega_2 \\ V_{G3x} \\ V_{G3y} \\ \omega_3 \end{matrix}$$

looking I ORD part of relationship found

We care also about y displacement

V_{G1y} is NOT relevant on equilibrium position



In fact moving as horizontal $V_{G1x}, V_{G2y} = 0$ NOT affect

Slider constraint, NO REL MOTION $\perp$ to leg and NOT rel rotation $\rightarrow$ rotates as the leg $\omega_3 = \dot{\theta}$

same rotation of both the legs

elongation of the springs: consider also (I), (II) ORD terms
 NOT needed for Rayrange eq. BUT we will need for MAX QUESTIONS
 ↑ to find the $[K_{el, II}]$

STATIC

$$\Delta l_1 = \Delta l_{01} - \frac{L_1}{2} \sin(\theta) + y \simeq \Delta l_{01} - \frac{L_1}{2} \theta + y$$

only x_{G1} influence

$$\Delta l_2 = \Delta l_{02} + x \quad (\text{Already Linear!})$$

only x affects Δl_2 dynamically

$$\Delta l_3 = \Delta l_{03} - (x_{G3} - x_{G30}) = \Delta l_{03} + L_1 \theta - (L_0 + x) \left(1 - \frac{\theta^2}{2}\right) - y + L_0$$

↓ from here
 slider @ equilibrium → $x_{G30} = L_0$

$$[\lambda_k] = \begin{array}{c|cc} & \theta & x & y \\ \hline \Delta l_1 & -L_1/2 & 0 & 1 \\ \hline \Delta l_2 & 0 & 1 & 0 \\ \hline \Delta l_3 & L_1 & -1 & -1 \end{array}$$

from IORD approx is easy to find also $K_{el, II, 2}$ and K_y from the x_G, y_G expressions...

↑ treat the leg of L bodies differently we have K_{g1}, K_{g2}, K_{g3} that we will sum later

$$[K_{g1}] = m_1 g \begin{array}{c|cc} & \theta & x & y \\ \hline & -L_1/2 & 0 & 0 \\ \hline & 0 & 0 & 0 \\ \hline & 0 & 0 & 0 \end{array}$$

3x3 SYMM MATRIX

2nd ord derivatives of y_{G1} to look at No dependence on x, y While $-L_1/2$ respect $(\theta^2/2)$

$$[K_{g2}] = m_2 g \begin{array}{c|cc} & \theta & x & y \\ \hline & -L_1 & 0 & 0 \\ \hline & 0 & 0 & 0 \\ \hline & 0 & 0 & 0 \end{array}$$

looky @ y_{G2}

$$[K_{g3}] = m_3 g \begin{bmatrix} -L_1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{looking at } y_{G3}$$

$(1 \cdot x \cdot g)$ term on y_{G3}

We can find $[K_{elII}]$

$$[K_{elII,1}] = [0]_{3 \times 3} \quad \text{NO 2^{nd} OR D term on } \Delta e_1$$

$$[K_{elII,2}] = [0]_{3 \times 3} \quad \text{linear } \Delta e_2$$

$$[K_{elII,3}] = K_3 \Delta l_{03} \begin{bmatrix} L_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{from II OR D terms on } \Delta e_3$$

$$[K] = \sum [K_{el}] + \sum [K_g] + [\lambda_k]^T [K_{ph}] [\lambda_k] \quad \text{overall}$$

$$[C] = [\lambda_c]^T [C_{ph}] [\lambda_c] \quad \text{with } [\lambda_c] \text{ equals to } [\lambda_k] \text{ but as } c_i \text{ instead of } k_i$$

VIRTUAL WORK

$$\delta W = \delta y_F^T (\underline{f}_{ph}) \quad \text{we have 2 excitation coming from } y, f(x)$$

$$[\lambda_F] = \begin{bmatrix} -L_1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} \delta x \\ \delta y \end{matrix}$$

$\underline{f}_{ph}(t) = \begin{Bmatrix} f(t) \\ H_0 \end{Bmatrix}$

$\delta y_F = \begin{Bmatrix} \delta x_{G3} \\ \delta y \end{Bmatrix}$

linearly link

only related to its variation of displacement δy

2 rows related to virtual variation absolute of G_3 and virtual displacement of 0

same as $[\lambda_{lm}]$ row of V_{G3}

We have all terms to obtain $[M], [K], [C], (\underline{f})^T = [\lambda_F]^T \cdot \underline{f}_{ph}$ right side

once found LAGRANGE $\rightarrow$ compute $W_i, \underline{X}$ as eg Val/vec problem
for $[M]^T [K]$ (MATLAB)

• compute FRF on $\Omega \in (0, 5) \text{ Hz}$ for cycle range to consider

• apply an force horiz external on m_3 $f_0 e^{i\Omega t}$ with $\Omega = 2\pi f$ $f \in (0, 5) \text{ Hz}$

• OUTPUT as $\Theta = X(1)$ first indip coord $\rightarrow$ obs, phase BOE

• force F_{el3} ? transmitted by $K_3 // C_3$

by extracting $X(1), X(2), X(3)$

what we need for FRF

(we find) $\Delta l_3 = L_1 X(1) - X(2) - X(3)$ FIRST ORDER APPROX

and find $F_3 = K \Delta l_3 + C \dot{\Delta l}_3 = K \Delta l_3 + i \Omega C \Delta l_3$ all effects

elongation as force $\rightarrow$ FRF = JUST 1st ORDER terms

• QUESTION: compute static pre-load of K_1 spring

(NOT provided before because $[K]_{el II, I} = 0$ does NOT affect $[K]$ Matrix but we want to compute it NOW!

(as done on previous exercise)

We do the same... 2 WAYS $\rightarrow$ 1) equilibrium equation (D'Alemb.) to solve

2) stability $\partial V = 0$ theorem, STATIONARY

$\downarrow$
total V is stationary at equilibrium

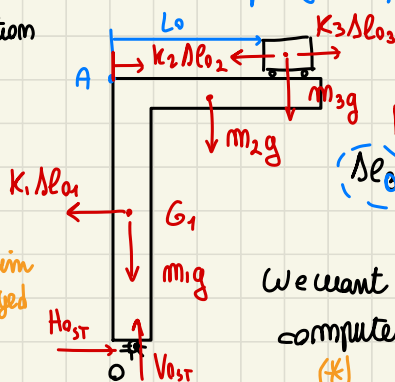
• METHOD 1 we are

referring to STATIC condition

of the system! we don't care about any term related to inertia or damper forces! STATIC

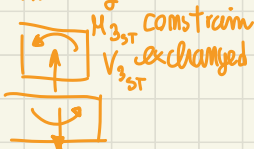
@ equilibrium position

caring only about STATIC terms!
Weight, spring forces



F_{el2} is INTERNAL
 Δl_{03} as part of problem Data
KNOWN from DATA

UNKNOWN neither invariance



We want one equation to compute $K_1 \Delta l_{01}$ without invariance constraints (K)

$\Delta \ell_{03}$ known with all constraints and $\Delta \ell_{01}, \Delta \ell_{02}$?

We have 6 UNKN $\rightarrow$ 6 EQ indep... as $\sum F_x / F_y / M = 0$

for body 2/2 / entire syst

equation

$\rightarrow$ 3 eq in total but some depends on others
there are 6 indep

HARD to
solve 6 eq...

easy simple solution $\Rightarrow$ one eliminating all constraints
providing directly $K_1 \Delta \ell_{01}$

considering all syst

so $V_3, M_3, K_2 \Delta \ell_{02}$

ONE INTERNAL negligible!

$$\sum_{\text{SYST}} \vec{M}_0 = 0$$

eliminate
unkn effect

$\rightarrow$
combination
of moment

$$+ K_1 \Delta \ell_{01} \cdot \frac{L_1}{2} - K_3 \Delta \ell_{03} L_1 + 0 - m_2 g \frac{L_2}{2} - m_3 g L_0 = 0$$

$\downarrow$
from here
you can find $\Delta \ell_{03}$

$\uparrow$
 $m_1 g$

on static
position
slider has L_0
arm

METHOD 2 through STATIONARITY of potential energy

counting down total V

$$V = m_1 g (h_1)^{y_{01}} + m_2 g (h_2)^{y_{02}} + m_3 g (h_3)^{y_{03}} + \frac{1}{2} K_1 \Delta \ell_1^2 + \frac{1}{2} K_2 \Delta \ell_2^2 + \frac{1}{2} K_3 \Delta \ell_3^2$$

$$V = m_1 g \frac{L_1}{2} \cos(\theta) + m_2 g (L_1 \cos(\theta) + \frac{L_2}{2} \sin(\theta)) + m_3 g (L_1 \cos(\theta) + (L_0 + x) \sin(\theta)) + \frac{1}{2} K_1 (\Delta \ell_{01} - \frac{L_1}{2} \sin(\theta) + y)^2 + \frac{1}{2} K_2 (\Delta \ell_{02} + x)^2 + \frac{1}{2} K_3 (\Delta \ell_{03} + L_1 \sin(\theta) - (L_0 + x) \cos(\theta) - y + L_0)^2$$

$\uparrow$
complete potential energy effect, including $\Delta \ell_0$ pre-load

(found previously for this)

STATIONARITY of Pot en @ eq. means $\partial V / \partial \theta, \partial V / \partial x, \partial V / \partial y \big|_{eq} = 0$

$\Downarrow$

$$\frac{\partial V}{\partial \theta} \bigg|_{\substack{\theta=0 \\ x=0 \\ y=0}} = 0$$

hard to compute $\rightarrow$ it is just mathematic
overall...

approach of derivation

on other problems identify method 1 can be HARD to find the best choice of the equations $\rightarrow$ so sometimes easier with STATIONARITY

$$\left. \frac{\partial V}{\partial \theta} \right|_{\substack{\theta=0 \\ x=0 \\ y=0}} = m_1 g \frac{L_1}{2} \cancel{(-\sin(\theta))}^{\theta=0} - m_2 g L_1 \cancel{\sin(\theta)} + m_2 g \frac{L_2}{2} \cancel{\cos(\theta)}^1 - m_3 g \cancel{L_1 \sin(\theta)} \\ + (L_0 + \cancel{x}) \cancel{\cos(\theta)}^1 m_3 g + k_1 \left(\Delta l_{01} - \frac{L_1}{2} \cancel{\sin(\theta)} + y \right) \cdot \left(-\frac{L_1}{2} \cancel{\cos(\theta)}^1 \right) \\ + k_3 \left(\Delta l_{03} + L_1 \cancel{\sin(\theta)} - (L_0 + \cancel{x}) \cancel{\cos(\theta)}^1 - y + L_0 \right) \left(L_1 \cancel{\cos(\theta)}^1 + (L_0 + \cancel{x}) \cancel{\sin(\theta)} \right) = 0$$

we should evaluate it @ $\left. \begin{matrix} \theta=0 \\ x=0 \\ y=0 \end{matrix} \right\}$ equilibrium

$\left\{ \begin{matrix} \cos \theta = 1 \\ \sin \theta = 0 \end{matrix} \right\} @ \theta = 0$ $\uparrow$ STATIONARY OF V Respect θ @ equilibrium

simplification
of equation...

$$= m_2 g \frac{L_2}{2} + m_3 g L_0 - k_1 \Delta l_{01} \frac{L_1}{2} + k_3 \Delta l_{03} L_1 = 0$$

EQUAL to the previously found from method 1
SAME RESULT

(decide which is best approach depending on complexity)

Involving only V or METHOD 2 we automatically neglect constraint forces! directly ignored by the kind of approach